

JESS Thermodynamic Database v8.9

Chemical species in reactions with Radium

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
Acetic-1 Acetate ion				
-1	71-50-1	558	C(2)H(3)O(2)	59.0446
Asp-2 Aspartate ion; Aminosuccinate ion; Asparagat ion; 2-Aminobutanedioate ion; L-2-Aminobutanedioate ion				
-2	63-71-8	776	C(4)H(5)N(1)O(4)	131.088
Benzoic-1 Benzoate ion; Benzenecarboxylate ion				
-1	766-76-7	208	C(7)H(5)O(2)	121.116
Br-1 Bromide ion				
-1	24959-67-9	1213	Br(1)	79.9040
C(s) Carbon; Graphite; Carbon, hexagonal				
0	7440-44-0	530	C(1)	12.0110
Cat-2 Catecholate ion; 1,2-Dihydroxybenzoate ion; 1,2-Benzenedioate ion; Pyrocatecholate ion; Oxyphenoxide				
-2	19021-48-8	333	C(6)H(4)O(2)	108.097
CDTA-4 trans-1,2-cyclohexylenedinitrilotetraacetate ion; trans-1,2-diaminocyclohexanetetraacetate ion; CDTA ion				
-4	22005-54-5	301	C(14)H(18)N(2)O(8)	342.306
Citric-3 Citrate ion				
-3	126-44-3	1347	C(6)H(5)O(7)	189.102

Cl-1 Chloride ion				
-1	16887-00-6	2242	Cl (1)	35.4530
Cl2 (g) Chlorine gas				
0	7782-50-5	589	Cl (2)	70.9060
CN-1 Cyanide ion				
-1	57-12-5	658	C (1) N (1)	26.0177
CO3-2 Carbonate ion				
-2	3812-32-6	1435	C (1) O (3)	60.0092
CrO4-2 Chromate ion				
-2	11104-59-9	127	Cr (1) O (4)	115.994
DTPA-5 Diethylenetrinitrilotetraacetate ion; 1,4,7-triazaheptane-1,1,7,7-pentaacetate ion; N,N-bis[2-[bis(carboxymethyl)amino]ethyl]glycinate ion; DTPA ion				
-5	---	343	C (14) H (18) N (3) O (10)	388.311
e-1 Electron				
-1	---	3399	E (1)	0.000000
EDTA-4 Ethylenediaminetetraacetate anion				
-4	---	1457	C (10) H (12) N (2) O (8)	288.214
EDTMP-8 Ethylenediaminetetramethylphosphonate ion				
-8	---	134	C (6) H (12) N (2) O (12) P (4)	428.064
EGTA-4 Ethylenebis(oxyethylenenitrilo)tetraacetate ion; EGTA ion				
-4	370-65-0	132	C (14) H (20) N (2) O (10)	376.320

EtDiAm Ethylenediamine; 1,2-Ethanediamine; 1,2-Diaminoethane; en !				
0	107-15-3	879	C(2)H(8)N(2)	60.0989
F-1 Fluoride ion				
-1	16984-48-8	1090	F(1)	18.9984
Fumaric-2 Fumarate ion; trans-Butenedioate ion				
-2	142-42-7	70	C(4)H(2)O(4)	114.058
Gly-1 Glycinate ion; Aminoacetate ion				
-1	---	1356	C(2)H(4)N(1)O(2)	74.0593
Glycolic-1 Glycolate ion				
-1	57122-18-6	342	C(2)H(3)O(3)	75.0440
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H(1)	1.00794
H+1_Citric-3				
-2	---	93	C(6)H(6)O(7)	190.109
H+1_CO3-2				
-1	---	253	C(1)H(1)O(3)	61.0171
H+1_EDTA-4				
-3	---	70	C(10)H(13)N(2)O(8)	289.222
H+1_PO4-3 Hydrogenphosphate ion				
-2	---	317	H(1)O(4)P(1)	95.9795
H+1_SulfSal-3				
-2	---	54	C(7)H(4)O(6)S(1)	216.165

H+1 (2) _PO4-3 Dihydrogenphospate ion				
-1	14066-20-7	284	H(2)O(4)P(1)	96.9875
H2 (g) Hydrogen gas; Hydrogen				
0	1333-74-0	1564	H(2)	2.01588
H2O Water				
0	7732-18-5	5947	H(2)O(1)	18.0153
HOEDTA-3 N-[2-[bis(carboxymethyl)amino]ethyl]-N-(2-hydroxyethyl)glycinate ion; N-(2-Hydroxyethyl)ethylenedinitrilotriacetate; HEDTA-3				
-3	14047-40-6	321	C(10)H(15)N(2)O(7)	275.238
I-1 Iodide ion				
-1	20461-54-5	894	I(1)	126.900
I2 (s) Iodine; Iodine, orthorhombic				
0	7553-56-2	244	I(2)	253.800
IO3-1 Iodate ion				
-1	15454-31-6	291	I(1)O(3)	174.898
Malic-2 Malate ion				
-2	149-61-1	682	C(4)H(4)O(5)	132.073
N2 (g) Nitrogen gas; Nitrogen				
0	7727-37-9	566	N(2)	28.0134
NH3 Ammonia, aqueous				
0	---	1462	H(3)N(1)	17.0305

NO3-1 Nitrate ion				
-1	14797-55-8	573	N(1)O(3)	62.0049
NTA-3 Nitrilotriacetate ion; N,N-bis(carboxymethyl)glycinate ion; Triglycollamate ion; Triglycinate ion; NTA ion				
-3	28528-44-1	751	C(6)H(6)N(1)O(6)	188.117
O2 (g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O(2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H(1)O(1)	17.0073
OxAcet-2 Oxalacetate ion; Oxaloacetate ion; Oxobutanedioate ion				
-2	---	43	C(4)H(2)O(5)	130.057
Oxalic-2 Oxalate ion				
-2	338-70-5	1241	C(2)O(4)	88.0196
Phenol-1 Phenolate ion; Hydroxybenzoate ion				
-1	3229-70-7	154	C(6)H(5)O(1)	93.1051
Pyruvic-1 Pyruvate ion				
-1	57-60-3	296	C(3)H(3)O(3)	87.0550
Ra+2 Radium(II) ion				
2	22541-36-2	131	Ra(1)	226.025
Ra+2_Acetic-1				
1	---	1	C(2)H(3)O(2)Ra(1)	285.070

Ra+2_Acetic-1(2)				
0	---	1	C(4)H(6)O(4)Ra(1)	344.114
Ra+2_Acetic-1(3)				
-1	---	1	C(6)H(9)O(6)Ra(1)	403.159
Ra+2_Asp-2				
0	---	1	C(4)H(5)N(1)O(4)Ra(1)	357.113
Ra+2_Benzoic-1				
1	---	1	C(7)H(5)O(2)Ra(1)	347.141
Ra+2_Benzoic-1(2)				
0	---	1	C(14)H(10)O(4)Ra(1)	468.256
Ra+2_Benzoic-1(3)				
-1	---	1	C(21)H(15)O(6)Ra(1)	589.372
Ra+2_Br-1				
1	---	1	Br(1)Ra(1)	305.929
Ra+2_Br-1(2)				
0	---	1	Br(2)Ra(1)	385.833
Ra+2_Br-1(3)				
-1	---	1	Br(3)Ra(1)	465.737
Ra+2_Br-1(4)				
-2	---	1	Br(4)Ra(1)	545.641
Ra+2_Br-1(5)				
-3	---	1	Br(5)Ra(1)	625.545
Ra+2_Br-1(6)				
-4	---	1	Br(6)Ra(1)	705.449

Ra+2_Cat-2				
0	---	1	C (6) H (4) O (2) Ra (1)	334.122
Ra+2_Cat-2 (2)				
-2	---	1	C (12) H (8) O (4) Ra (1)	442.218
Ra+2_Cat-2 (3)				
-4	---	1	C (18) H (12) O (6) Ra (1)	550.315
Ra+2_CDTA-4				
-2	---	1	C (14) H (18) N (2) O (8) Ra (1)	568.331
Ra+2_Citric-3				
-1	---	1	C (6) H (5) O (7) Ra (1)	415.127
Ra+2_Cl-1				
1	---	2	Cl (1) Ra (1)	261.478
Ra+2_Cl-1 (2)				
0	---	2	Cl (2) Ra (1)	296.931
Ra+2_Cl-1 (2) _ (s) Radium chloride				
0	10025-66-8	1	Cl (2) Ra (1)	296.931
Ra+2_Cl-1 (2) _H2O (2) _ (s)				
0	---	2	Cl (2) H (4) O (2) Ra (1)	332.962
Ra+2_Cl-1 (3)				
-1	---	1	Cl (3) Ra (1)	332.384
Ra+2_Cl-1 (4)				
-2	---	1	Cl (4) Ra (1)	367.837
Ra+2_Cl-1 (5)				
-3	---	1	Cl (5) Ra (1)	403.290

Ra+2_Cl-1 (6)				
-4	---	1	Cl (6) Ra (1)	438.743
Ra+2_CN-1				
1	---	1	C (1) N (1) Ra (1)	252.043
Ra+2_CN-1 (2)				
0	---	1	C (2) N (2) Ra (1)	278.060
Ra+2_CN-1 (3)				
-1	---	1	C (3) N (3) Ra (1)	304.078
Ra+2_CO3-2				
0	---	2	C (1) O (3) Ra (1)	286.034
Ra+2_CO3-2 (s) Radium carbonate				
0	---	2	C (1) O (3) Ra (1)	286.034
Ra+2_CO3-2 (2)				
-2	---	1	C (2) O (6) Ra (1)	346.043
Ra+2_CO3-2 (3)				
-4	---	1	C (3) O (9) Ra (1)	406.053
Ra+2_CO3-2 (4)				
-6	---	1	C (4) O (12) Ra (1)	466.062
Ra+2_CO3-2 (5)				
-8	---	1	C (5) O (15) Ra (1)	526.071
Ra+2_CrO4-2				
0	---	1	Cr (1) O (4) Ra (1)	342.019
Ra+2_DTPA-5				
-3	---	1	C (14) H (18) N (3) O (10) Ra (1)	614.336

Ra+2_EDTA-4				
-2	---	2	C (10) H (12) N (2) O (8) Ra (1)	514.239
Ra+2_EDTMP-8				
-6	---	1	C (6) H (12) N (2) O (12) P (4) Ra (1)	654.089
Ra+2_EGTA-4				
-2	---	1	C (14) H (20) N (2) O (10) Ra (1)	602.345
Ra+2_EtDiAm				
2	---	1	C (2) H (8) N (2) Ra (1)	286.124
Ra+2_EtDiAm (2)				
2	---	1	C (4) H (16) N (4) Ra (1)	346.223
Ra+2_F-1				
1	---	1	F (1) Ra (1)	245.023
Ra+2_F-1 (2)				
0	---	1	F (2) Ra (1)	264.022
Ra+2_F-1 (3)				
-1	---	1	F (3) Ra (1)	283.020
Ra+2_F-1 (4)				
-2	---	1	F (4) Ra (1)	302.019
Ra+2_F-1 (5)				
-3	---	1	F (5) Ra (1)	321.017
Ra+2_F-1 (6)				
-4	---	1	F (6) Ra (1)	340.015
Ra+2_Fumaric-2				
0	---	1	C (4) H (2) O (4) Ra (1)	340.083

Ra+2_Gly-1				
1	---	1	C (2) H (4) N (1) O (2) Ra (1)	300.084
Ra+2_Gly-1 (2)				
0	---	1	C (4) H (8) N (2) O (4) Ra (1)	374.144
Ra+2_Gly-1 (3)				
-1	---	1	C (6) H (12) N (3) O (6) Ra (1)	448.203
Ra+2_Glycolic-1				
1	---	1	C (2) H (3) O (3) Ra (1)	301.069
Ra+2_Glycolic-1 (2)				
0	---	1	C (4) H (6) O (6) Ra (1)	376.113
Ra+2_Glycolic-1 (3)				
-1	---	1	C (6) H (9) O (9) Ra (1)	451.157
Ra+2_H+1_Citric-3				
0	---	1	C (6) H (6) O (7) Ra (1)	416.134
Ra+2_H+1_CO3-2				
1	---	1	C (1) H (1) O (3) Ra (1)	287.042
Ra+2_H+1_PO4-3				
0	---	1	H (1) O (4) P (1) Ra (1)	322.005
Ra+2_H+1_SulfSal-3				
0	---	1	C (7) H (4) O (6) Ra (1) S (1)	442.190
Ra+2_H+1 (10)_PO4-3 (5)				
-3	---	1	H (10) O (20) P (5) Ra (1)	710.962
Ra+2_H+1 (12)_PO4-3 (6)				
-4	---	1	H (12) O (24) P (6) Ra (1)	807.950

Ra+2_H+1 (2)_CO3-2 (2)				
0	---	1	C (2) H (2) O (6) Ra (1)	348.059
Ra+2_H+1 (2)_PO4-3				
1	---	1	H (2) O (4) P (1) Ra (1)	323.013
Ra+2_H+1 (2)_PO4-3 (2)				
-2	---	1	H (2) O (8) P (2) Ra (1)	417.984
Ra+2_H+1 (3)_CO3-2 (3)				
-1	---	1	C (3) H (3) O (9) Ra (1)	409.076
Ra+2_H+1 (3)_PO4-3 (3)				
-4	---	1	H (3) O (12) P (3) Ra (1)	513.964
Ra+2_H+1 (4)_PO4-3 (2)				
0	---	1	H (4) O (8) P (2) Ra (1)	420.000
Ra+2_H+1 (4)_PO4-3 (4)				
-6	---	1	H (4) O (16) P (4) Ra (1)	609.943
Ra+2_H+1 (5)_PO4-3 (5)				
-8	---	1	H (5) O (20) P (5) Ra (1)	705.923
Ra+2_H+1 (6)_PO4-3 (3)				
-1	---	1	H (6) O (12) P (3) Ra (1)	516.987
Ra+2_H+1 (6)_PO4-3 (6)				
-10	---	1	H (6) O (24) P (6) Ra (1)	801.902
Ra+2_H+1 (8)_PO4-3 (4)				
-2	---	1	H (8) O (16) P (4) Ra (1)	613.975
Ra+2_HOEDTA-3				
-1	---	1	C (10) H (15) N (2) O (7) Ra (1)	501.263

Ra+2_I-1				
1	---	1	I (1) Ra (1)	352.925
Ra+2_I-1 (2)				
0	---	1	I (2) Ra (1)	479.825
Ra+2_I-1 (3)				
-1	---	1	I (3) Ra (1)	606.725
Ra+2_I-1 (4)				
-2	---	1	I (4) Ra (1)	733.625
Ra+2_I-1 (5)				
-3	---	1	I (5) Ra (1)	860.525
Ra+2_I-1 (6)				
-4	---	1	I (6) Ra (1)	987.425
Ra+2_IO3-1				
1	---	1	I (1) O (3) Ra (1)	400.923
Ra+2_IO3-1 (2)				
0	---	1	I (2) O (6) Ra (1)	575.821
Ra+2_IO3-1 (2)_(s)				
0	---	1	I (2) O (6) Ra (1)	575.821
Ra+2_IO3-1 (3)				
-1	---	1	I (3) O (9) Ra (1)	750.720
Ra+2_Malic-2				
0	---	1	C (4) H (4) O (5) Ra (1)	358.098
Ra+2_NH3				
2	---	1	H (3) N (1) Ra (1)	243.056

Ra+2_NH3(2)				
2	---	1	H(6)N(2)Ra(1)	260.086
Ra+2_NO3-1				
1	---	1	N(1)O(3)Ra(1)	288.030
Ra+2_NO3-1(2)				
0	---	2	N(2)O(6)Ra(1)	350.035
Ra+2_NO3-1(2)_(s) Radium nitrate				
0	10213-12-4	2	N(2)O(6)Ra(1)	350.035
Ra+2_NO3-1(3)				
-1	---	1	N(3)O(9)Ra(1)	412.040
Ra+2_NTA-3				
-1	---	1	C(6)H(6)N(1)O(6)Ra(1)	414.142
Ra+2_OH-1				
1	---	5	H(1)O(1)Ra(1)	243.032
Ra+2_OH-1(2)				
0	---	1	H(2)O(2)Ra(1)	260.040
Ra+2_OH-1(2)_(s) Radium hydroxide				
0	---	1	H(2)O(2)Ra(1)	260.040
Ra+2_OH-1(2)_H2O(8)_(s)				
0	---	1	H(18)O(10)Ra(1)	404.162
Ra+2_OH-1(3)				
-1	---	1	H(3)O(3)Ra(1)	277.047
Ra+2_OH-1(4)				
-2	---	1	H(4)O(4)Ra(1)	294.054

Ra+2_OxAcet-2				
0	---	1	C (4) H (2) O (5) Ra (1)	356.082
Ra+2_Oxalic-2				
0	---	1	C (2) O (4) Ra (1)	314.045
Ra+2_Oxalic-2 (2)				
-2	---	1	C (4) O (8) Ra (1)	402.064
Ra+2_Phenol-1				
1	---	1	C (6) H (5) O (1) Ra (1)	319.130
Ra+2_Phenol-1 (2)				
0	---	1	C (12) H (10) O (2) Ra (1)	412.235
Ra+2_Phenol-1 (3)				
-1	---	1	C (18) H (15) O (3) Ra (1)	505.340
Ra+2_Pyruvic-1				
1	---	1	C (3) H (3) O (3) Ra (1)	313.080
Ra+2_SCN-1				
1	---	1	C (1) N (1) Ra (1) S (1)	284.103
Ra+2_SCN-1 (2)				
0	---	1	C (2) N (2) Ra (1) S (2)	342.180
Ra+2_SCN-1 (3)				
-1	---	1	C (3) N (3) Ra (1) S (3)	400.258
Ra+2_SO4-2				
0	---	2	O (4) Ra (1) S (1)	322.083
Ra+2_SO4-2_(s) Radium sulphate				
0	7446-16-4	2	O (4) Ra (1) S (1)	322.083

Ra+2_SO4-2 (2)				
-2	---	1	O (8) Ra (1) S (2)	418.140
Ra+2_SO4-2 (3)				
-4	---	1	O (12) Ra (1) S (3)	514.198
Ra+2_SO4-2 (4)				
-6	---	1	O (16) Ra (1) S (4)	610.255
Ra+2_SO4-2 (5)				
-8	---	1	O (20) Ra (1) S (5)	706.313
Ra+2_Succinic-2				
0	---	1	C (4) H (4) O (4) Ra (1)	342.098
Ra+2_Tartaric-2				
0	---	1	C (4) H (4) O (6) Ra (1)	374.097
Ra (g) Radium gas				
0	---	1	Ra (1)	226.025
Ra (s) Radium; Radium white, cubic				
0	7440-14-4	18	Ra (1)	226.025
RaH2 (s) Radium hydride				
0	---	1	H (2) Ra (1)	228.041
RaO (s)				
0	---	2	O (1) Ra (1)	242.024
S (s) Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb				
0	7704-34-9	619	S (1)	32.0600

SCN-1 Thiocyanate ion; NCS-1 equivalent				
-1	302-04-5	785	C(1)N(1)S(1)	58.0777
SO4-2 Sulfate ion; Sulphate ion				
-2	14808-79-8	1270	O(4)S(1)	96.0576
Succinic-2 Succinate ion				
-2	56-14-4	635	C(4)H(4)O(4)	116.073
Tartaric-2 Tartrate ion				
-2	---	393	C(4)H(4)O(6)	148.072